Optimizing Customer Spending in TokoBli E-commerce: A Significant Factor Analysis

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Dataset

- https://docs.google.com/spreadsheets/d/1PV56d7D2pa5JOVa1dNzGaQb2lt9ebnQd/edit?gid=1512973310#gid=1512973310

1. Which variables have a **strong relationship** with each other? (Hint: relation = **correlation**)

Correlation Matrix

	Age	Income	Tenure	Avg. Session Time	Total Promo	Bounce Rate	Yearly Spending
Age	1,00						
Income	0,14	1,00					
Tenure	0,16	0,38	1,00				
Avg. Session Time	0,19	0,48	0,69	1,00			
Total Promo	0,21	0,45	0,62	0,78	1,00		
Bounce Rate	-0,13	-0,45	-0,57	-0,79	-0,79	1,00	
Yearly Spending	0,22	0,53	0,71	0,91	0,88	-0,86	1,00

	positive	very strong	0,7 - 1
	positive	strong	0,5 - 0,7
	positive	moderate	0,3 - 0,5
	positive	weak	0 - 0,3
	negative	weak	-0,3 - 0
	negative	moderate	-0,5 - -0,3
	negative	strong	-0,7 - -0,5
	negative	very strong	-1 - -0,7

Positive – Very Strong	**Positive - Strong**
Tenure --- Yearly Spending	Income --- Yearly Spending
Average Session Time --- Total Promo	Tenure --- Average Session Time
Average Session Time --- Yearly Spending	Tenure --- Total Promo
Total Promo – Yearly Spending	

2. Correlation insights

- **Tenure, Average Session Time, & Total Promo** have a **very strong** relationship with **Yearly Spending**

- **Income** has a strong relationship with **Yearly Spending**

- **Tenure** has a strong relationship with **Average Session Time & Total Promo**

3. Escalate the Correlation analysis into regression

- Tenure, Average Session Time, & Total Promo have a very strong relationship with Yearly Spending
- Income has a strong relationship with Yearly Spending
- But statistical relationship does not always imply a causality.
- In order to determine causality & true factors which have a significant effect on the amount of yearly spending from customers, we have to do regression analysis

4. What variable has a **significant impact** on the Yearly Spending? (Hint: **impact = regression**)

Step by step Regression Building :

1. Problem definition :

   What variable has a significant impact on the Yearly Spending?

2. Analyse data :
   - Based on correlation analysis, we choose variable which have strong & very strong relationship with Yearly Spending to be Independent variable (X).
   - Dependent variable (Y) = Yearly Spending
   - Independent variable (X) = Tenure, Average Session Time, Total Promo, & Income

3. Model selection :

   Linear Regression

   For numeric dependent variable

Model Fitting

Regression Statistics	
Multiple R	0,956162756
R Square	91,42%
Adjusted R Square	91,31%
Standard Error	1519247,182
Observations	299

Adjusted R Square = 91% → 91% Yearly Spending is influenced by Tenure, Average Session Time, Total Promo, & Income

ANOVA					
	df	SS	MS	F	Significance F
Regression	4	7,23469E+15	1,80867E+15	783,6150329	0
Residual	294	6,78585E+14	2,30811E+12		
Total	298	7,91327E+15			

Simultant Test

- H0 = 0 independent variables influence dependent variable
- H1 = ≥1 independent variables influence dependent variable
- Alpha = 5%
- Significance F (0) < Alpha (0,05) → Reject H0, accept H1
- At least one independent variables influence dependent variable
- But, which one?

Model Fitting

	P-value			
Intercept	0	P-value < Alpha	H1 accepted, H0 rejected	
Income	0	P-value < Alpha	H1 accepted, H0 rejected	
Tenure	0	P-value < Alpha	H1 accepted, H0 rejected	
Avg. Session Time	0	P-value < Alpha	H1 accepted, H0 rejected	
Total Promo	0	P-value < Alpha	H1 accepted, H0 rejected	

Partial Test

- H0 = independent variable do not influence dependent variable
- H1 = independent variable influence dependent variable
- Alpha = 5%
- Intercept, Income, Tenure, Avg. Session Time, & Total Promo influence Yearly Spending

	Coefficients
Intercept	-694012,813
Income	0,043
Tenure	61305,723
Avg. Session Time	7747,515
Total Promo	184006,883

Regression Equation

Yearly Spending = -694012,813 + 0,043 * Income + 61305,723 * Tenure
+ 7747,515 * Avg. Session Time + 184006,883 * Total Promo

Model Evaluation

- MAE = 1081584,876
- MAPE = 17,47%

Assumption Test

	Income	Tenure	Avg. Session T	Total Promo
Income	1			
Tenure	0,38	1,00		
Avg. Session Time	0,48	0,69	1,00	
Total Promo	0,45	0,62	0,78	1

- Non Multicolinearity test
- No independent variable have correlation coefficient > 0,8 → **no correlation between independent variables**
- **No independent variables need to be removed**

5. Find the best regression = **stepwise regression**

Iterasi 1		
	Coefficients	P-value
Intercept	1947879,706	0,00
Age	30555,49209	0,07
Income	0,035824641	0,00
Tenure	64158,04723	0,00
Avg. Session Time	6286,982772	0,00
Total Promo	133215,6971	0,00
Bounce Rate	-50349,9385	0,00

Iterasi 2		
	Coefficients	P-value
Intercept	2590241,490	0,00
Income	0,037	0,00
Tenure	64576,213	0,00
Avg. Session Time	6341,902	0,00
Total Promo	136147,724	0,00
Bounce Rate	-49209,126	0,00

Regression Equation

Yearly Spending = 2590241,490 + 0,037 * Income + 64576,213 * Tenure + 6341,901 * Avg. Session + 136147,724 * Total Promo -49209,126 * Bounce Rate

6. Predict the Potential yearly spending of a potential new customer who on-boarded about 1 month ago

- Based on the following characteristics:

Age	Income	Tenure	Avg. Session Time	Total Promo	Bounce Rate
20	20,000,000	1	1000	20	30

Yearly Spending = 2590241,490 + 0,037 * Income + 64576,213 * Tenure + 6341,901 * Avg. Session + 136147,724 * Total Promo -49209,126 * Bounce Rate

= 2590241,490 + 0,037 * 20000000 + 64576,213 * 1 + 6341,901 * 1000 + 136147,724 * 20 - 49209,126 * 30

= Rp10.983.399,40

7. Based all points what **insights recommendation(s)** can you provide?

Insights Factors with significant effect on yearly spending	**Business Impact**	**Isolation**	**Prioritization**	**Recommendation**
Income	Focusing on customer segment with high value	High income customer segment	High (High impact – High effort – Many data point)	1. Push premium product for high income customer 2. Upselling and cross selling for high income customer
Tenure & Bounce Rate	More retention more revenue	Retention	High (High impact – High effort – Many data point)	1. Loyality program 2. Develop better UI UX experience

Insights Factors with significant effect on yearly spending	**Business Impact**	**Isolation**	**Prioritization**	**Recommendation**
Total Promo	Promotion attract customer	Discount	High (High impact – High effort – Many data point)	1. Discount, voucher, cashback, promo variation
Avg. Session Time	More session time more revenue	Screen time	High (High impact – High effort – Many data point)	1. Interactive content inside apps (mini games) 2. Simple navigation 3. Personalize homepage 4. Community feature

Conclusion

- Factors which have a significant effect on the amount of yearly spending from customers are Income, Tenure, Avg. Session Time, Total Promo, & Bounce Rate
- Develop campaign and business strategies targetting customers with High Income, High Tenure, High Avg. Session Time, High Total Promo, & Low Bounce Rate